



Microbiological precipitation of CaCO_3

Shannon Stocks-Fischer, Johnna K. Galinat, Sookie S. Bang*

Department of Chemistry and Chemical Engineering, South Dakota School of Mines and Technology, 501 E. Saint Joseph Street, Rapid City, SD 57701, USA

Accepted 14 May 1999

Abstract

The process of microbial mineral plugging in porous media is common in nature. We examined physical and biochemical properties of CaCO_3 precipitation induced by *Bacillus pasteurii*, an alkalophilic soil microorganism. X-ray diffraction analysis quantified the composition of the mineral deposited in sand and identified the CaCO_3 crystal as calcite. Examination by scanning electron microscopy identified bacteria in the middle of calcite crystals, which acted as nucleation sites. The rate of microbiological CaCO_3 precipitation correlated with cell growth and was significantly faster than that of chemical precipitation. Biochemical properties of urease (urea amidohydrolase, E.C. 3.5.1.5) from *B. pasteurii* that was indirectly involved in CaCO_3 precipitation were examined to understand the kinetics of the microbiological process. Urease from *B. pasteurii* exhibited a relatively low affinity for urea at pH 7.0 with a K_m of 41.6 mM and $V_{\max}$ of $3.55 \text{ mM min}^{-1} \text{ mg}^{-1}$ protein and increased affinity at pH 7.7 with a K_m of 26.2 mM and $V_{\max}$ of $1.72 \text{ mM min}^{-1} \text{ mg}^{-1}$ protein. Results of kinetic studies indicate that urease activity and its affinity to urea are significantly high at the pH where calcite precipitation is favorable. Our findings further suggest a potential use of the microbial calcite precipitation process in remediation of the surface and subsurface of porous media. © 1999 Published by Elsevier Science Ltd. All rights reserved.

Keywords: Calcite precipitation; *Bacillus pasteurii*; Urease; Kinetics; Bioremediation

1. Introduction

Microbial metabolic activities often contribute to selective cementation by producing relatively insoluble organic and inorganic compounds intra- or extracellularly. Some microorganisms produce glycocalyx on the cell wall which remains in the natural environment even after cell death (Lappin-Scott et al., 1988; MacLeod et al., 1988), while others accumulate inorganic compounds such as phosphorites, carbonates, silicates and iron and manganese oxides (Beveridge et al., 1983; Ghiorse, 1984; Knoll, 1985; Ruiz et al., 1988; Rivadeneya et al., 1991). In a natural setting, precipitation processes continue at a slow rate over geological time, plugging cracks in highly permeable rock formations (Hart et al., 1960; Kantzas et al., 1992). The importance of selective cementation has been widely

recognized in Petroleum, Geological and Civil Engineering. It has been documented that cracks in rock formations, especially in oil reservoirs, could be remediated by microorganisms (Updegraff, 1982; Finnerty and Singer, 1983).

In our previous studies (Gollapudi et al., 1995; Zhong and Islam, 1995), *Bacillus pasteurii* induced a mineral plugging in surface fractures and fissures of granite. *B. pasteurii* uses urea as an energy source and produces ammonia which increases pH in the proximal environment, causing Ca^{2+} and CO_3^{2-} to precipitate as CaCO_3 (Kroll, 1990). The microbial process was found to be most effective in remediating fissures with an average width of 2.7 mm. Among the filling materials that were mixed with *B. pasteurii* for fissure remediation of granite, the silica (10%) and sand (90%) mixture produced the highest compressive strength and lowest permeability. Thus, the potential of this technique as a long-term remediation tool is highly feasible for cementation of various structural formations.

* Corresponding author.

Results of our studies further suggested that microbial plugging could be greatly enhanced using microorganisms with high urease enzyme activities indirectly involved in CaCO_3 consolidation.

We present physical and biochemical evidence of microbial mineral cementation, in which microbial metabolic activities play an important role. Microbial sedimentation of CaCO_3 on the surface and subsurface of sand columns was examined by scanning electron microscopy and X-ray diffraction quantitative analysis. The biochemical processes were investigated by comparing kinetics of microbiological and chemical CaCO_3 precipitation. Furthermore, urease activity and its Michaelis–Menten kinetics were evaluated at different pH values, which is a key factor in calcite precipitation.

2. Materials and methods

2.1. Microorganism and growth conditions

B. pasteurii ATCC 6453 was used throughout the study. Medium (Tris–YE) for stock and pilot cultures contained the following ingredients l^{-1} of glass distilled water: Tris–HCl, 130 mM (pH 9.0); $(\text{NH}_4)_2\text{SO}_4$, 10 g; and yeast extract, 20 g; to which 1.5% agar was added to obtain a solid medium for the stock culture. Individual ingredients were autoclaved separately and mixed afterward to avoid precipitation. CaCO_3 precipitation experiments were carried out in liquid medium (urea– CaCl_2) containing the following l^{-1} of glass distilled water: nutrient broth (Bacto), 3 g; urea, 20 g; NH_4Cl , 10 g; and NaHCO_3 , 2.12 g (equivalent to 25.2 mM). The pH of the medium was adjusted to 6.0 with 6 N HCl prior to autoclaving. 10 ml of filter-sterilized solution containing 1.40, 2.80 or 5.60 g CaCl_2 was added afterward to yield 12.6, 25.2 or 50.4 mM Ca^{2+} , respectively. The final pH of the medium was measured to be 8.0. *B. pasteurii* was grown at 30°C under aerobic conditions for stock and pilot cultures and at 25°C for CaCO_3 precipitation experiments. Broth cultures were incubated in a water bath shaker (Lab-line, Model 3540) operated at 200 rpm. Cell concentrations were determined by viable cell counting on Tris–YE plates.

2.2. Microbial plugging of sand column

Two sets of 600 ml of *B. pasteurii* were grown in Tris–YE medium until the cells reached the late exponential stage. Upon termination of the growth, one set of the culture was autoclaved at 121°C for 20 min to kill bacteria and endospores. When autoclaved cell suspension was plated on YE plates, no viable cells were counted. Each of the killed or live bacterial culture

was harvested ($5000 \times g$, 10 min, Sorvall, RC-2B) and washed with saline. Cells were suspended in urea– CaCl_2 medium and mixed with 100 g sterile sand. Sand slurry containing killed or live cells was packed into a 60-ml plastic syringe column. Separately, the same amount of sand without bacteria was also packed into a column. The three columns were fed continuously by gravity with urea– CaCl_2 medium containing 25.2 mM of CaCl_2 at room temperature to mimic the natural environmental conditions. When the flow stopped in the column packed with live bacteria in 10 d, the experiments in the three columns were terminated. After air-drying, the column samples were subjected to further analyses by X-ray diffraction and scanning electron microscopy.

2.3. X-ray diffraction (XRD) quantitative analysis

For XRD quantitative analysis of consolidated minerals, four different samples were prepared: a plugged sand column treated with bacteria and urea– CaCl_2 medium; a sand column treated with killed bacteria and urea– CaCl_2 medium; a sand column without bacteria but treated with medium and a sand sample treated without bacteria or medium. The sand column loaded with bacteria was so tightly plugged that the column was fractured with a mechanical knife. Samples from the treated columns were taken at depths extending from the surface to 0.5 cm. All samples, including one from the untreated sand, were individually ground and sieved through a 45 μm diameter mesh. Individual samples were scanned by X-ray diffraction (powder X-ray diffractometer, Philips Electronic Instruments) which analyzed the weight fractions of the individual chemical components (Davis et al., 1986).

2.4. Scanning electron microscopy (SEM)

Prior to microscopic examination, consolidated sand cores of the bacteria–sand column were cut open. Fractured surfaces were carbon-coated and examined with a JEOL JSM-840A scanning electron microscope at accelerating voltages ranging from 30 to 35 kV. Techniques of secondary electron imaging (SEI) and back-scattered electron imaging (BSEI) were employed for electron microscopy; the back-scattered imaging located bacteria with superior contrast.

2.5. Microbiologically-induced CaCO_3 precipitation

Microbial CaCO_3 precipitation experiments were carried out with three different concentrations of CaCl_2 : 12.6, 25.2 and 50.4 mM. At each concentration of CaCl_2 , duplicate sets of 10 50-ml Erlenmeyer flasks containing 10 ml of urea– CaCl_2 medium were pre-

pared. One replicate set was inoculated with live bacteria (1.0×10^6 cells ml^{-1}) and the other with an equivalent concentration of killed bacteria as control. The CaCO_3 precipitation experiment was carried out for 54 h with shaking. At 2-h intervals (starting from time zero) for the first 12 h and longer intervals afterward, one flask from each set was terminated. Immediately, an aliquot from the control flask was taken to determine the amount of NH_3 in the medium, while two separate aliquots from the flask with live cells were taken to determine the amount of NH_3 and to count cells, respectively. Then the broth of each flask was centrifuged to quantify the insoluble Ca^{2+} in the precipitates as well as the soluble Ca^{2+} in the supernatant separately. Ca^{2+} contents in both supernatant and precipitate were measured by the EDTA titration method (APHA, 1989). Ammonia liberated in the medium as a result of urea hydrolysis was detected by spectrophotometric assay (Natarajan, 1995). Due to the complexity of assay procedures at each interval, the complete set of each concentration of CaCl_2 was run twice instead of replicating flasks.

2.6. Chemical CaCO_3 precipitation at various pH values

The pH values of distilled water containing 25.2 mM HCO_3^- (control) and urea medium without CaCl_2 were adjusted with 1 N NaOH ranging from 8 to 13, respectively. Samples at each pH were prepared in triplicate. Then, 25.2 mM CaCl_2 was added to each sample of both control and urea medium to initiate CaCO_3 precipitation. After chemical precipitation proceeded for 12 h, each sample was centrifuged ($3000 \times g$, 10 min) to collect the insoluble CaCO_3 . The chemical precipitation experiments were carried out at room temperature. Precipitated CaCO_3 from each sample was measured by EDTA titration (APHA, 1989).

2.7. Preparation of the crude cell-free extract

Cells grown in 300 ml of Tris–YE medium were harvested at 4°C by centrifugation ($5000 \times g$, 10 min) and

washed twice in buffer containing sodium phosphate, 0.1 M (pH 7.0); EDTA, 1 mM; and 2-mercaptoethanol, 5 mM. The cell pellet was either used immediately or frozen at -20°C until needed. Cells were suspended in the buffer and disintegrated by ultrasonic treatment at the power output of 75 W (Sonics and Materials, VC 250) for 100 s on ice. After centrifugation at $13,000 \times g$ (Sorvall, RC-2B) to remove disrupted cells and particles, the supernatant was used as the crude cell-free extract for enzyme assay.

2.8. Assay of urease activity

Urease converts 1 mole of urea to 2 moles of ammonia and 1 mole of CO_2 . The urease activities in the cell extract and culture filtrate were assayed using the spectrophotometric method of Natarajan (1995). Absorbance at 626 nm (Hewlett Packard, Model 8452A) was recorded to quantify the ammonia produced upon enzymatic hydrolysis of urea. NH_4Cl (50 to 250 μM) was used as the standard. One unit of urease is defined as the amount of enzyme hydrolyzing 1 μmol urea min^{-1} . Protein concentration was determined by the modified procedure of Bradford (1976), using a Bio-Rad protein assay solution.

3. Results

3.1. XRD analysis of microbiologically-induced CaCO_3 precipitation

Table 1 summarizes the results of XRD quantitative analyses of four different sand samples. The most abundant compound was clearly quartz, the main component of sand. CaCO_3 crystals were identified as calcite, not aragonite which is more common in seawater or magnesium-rich aqueous solutions (Berner, 1975; Rivadeneyra et al., 1991). Calcite constituted 30.2% of the total weight of the column plugged by bacteria, but none was detected in column samples without live cells. In columns 1, 2 and 3 which were supplied with medium, urea was consumed by growing

Table 1
XRD quantitative analysis of the final weight fractions^a of sand samples^b

Sample	Quartz (SiO_2)	Calcite (CaCO_3)	Gehlenite ($\text{Ca}_2\text{Al}(\text{Al},\text{Si})_2\text{O}_7$)	Mullite ($\text{Al}_6\text{Si}_2\text{O}_{13}$)	Hematite (Fe_2O_3)	Goethite ($\alpha\text{-FeO}(\text{OH})$)	Urea ($\text{CH}_4\text{N}_2\text{O}$)
1	0.683 (0.065)	0.302 (0.066)	ND	0.015 (0.004)	trace	ND	ND
2	0.892 (0.015)	ND	0.083 (0.011)	0.012 (0.003)	trace	0.003 (0.001)	0.030 (0.005)
3	0.954 (0.009)	ND	ND	0.032 (0.008)	0.003 (0.001)	ND	0.011 (0.003)
4	0.960 (0.010)	ND	ND	0.039 (0.010)	trace	ND	ND

^a Numbers represent an average of weight fraction values obtained from energy-dispersive XRD quantitative analysis; (), variance errors; trace, <0.001 ; ND, not detected.

^b Sample 1, sand treated with *B. pasteurii* and medium; sample 2, sand treated with killed *B. pasteurii* and medium; sample 3, sand treated with medium; sample 4, untreated sand.

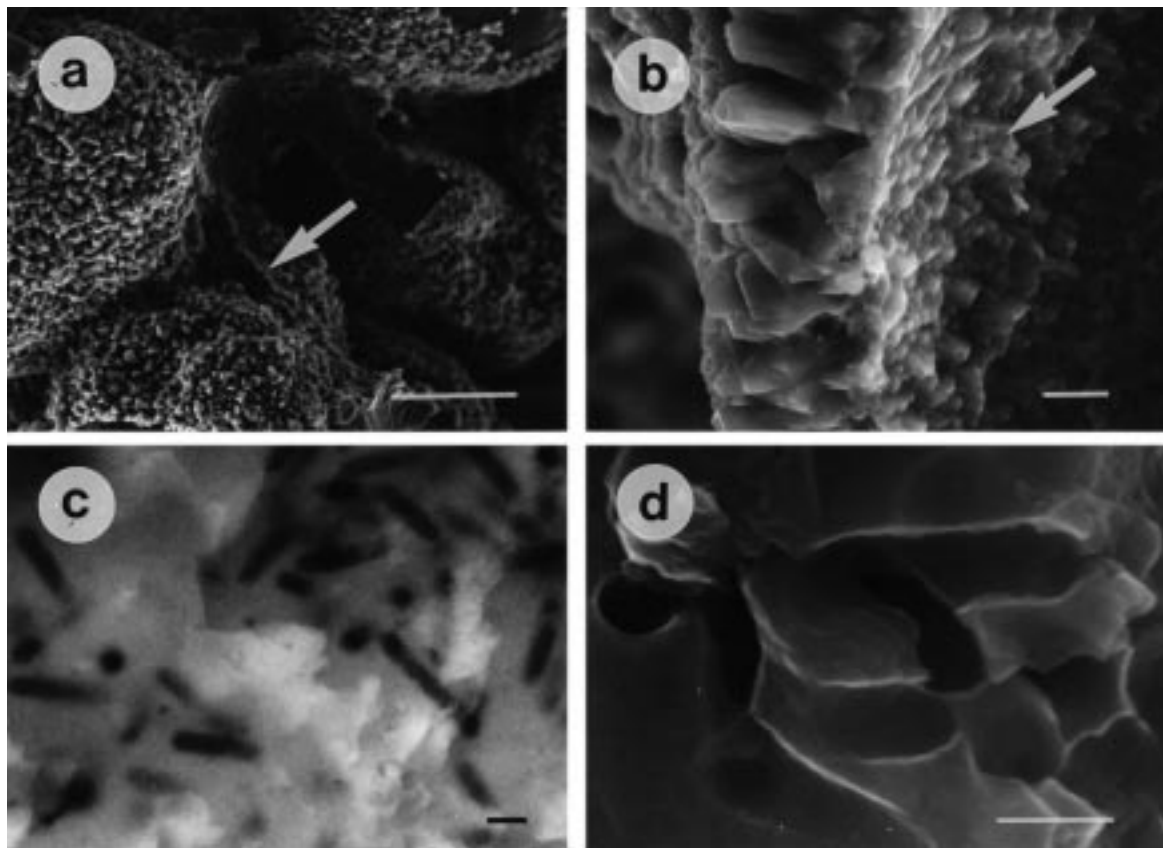


Fig. 1. Scanning electron micrographs showing bacteria embedded in the crystals of cemented material within sand columns. Electron micrography employed two types of imaging techniques: secondary electron imaging (SEI) and back-scattered electron imaging (BSEI). (a) SEI. Calcite crystals have formed over the sand particles in the column fracture. Bar, 100 μm . (b) SEI, details of crystal formations similar to the area marked with an arrow in (a). At this magnification, the fine structure of the crystallites is easily discerned and evident. Bar, 10 μm . (c) BSEI. Magnification of the area marked with an arrow in (b). Dark cylindrical images represent *B. pasteurii* embedded in the crystals. Bar, 1 μm . (d) SEI. Enlarged image of a section in (c). Empty tunnel and sphere shapes represent the spaces once occupied by bacilli and endospores. Bar, 1 μm .

bacteria in column 1, while unmetabolized urea was detected in columns 2 and 3 containing killed and no bacteria, respectively. It is noteworthy that a small amount of Ca^{2+} (8.3%) was crystallized as a form of gehlenite in the presence of killed bacteria. However, it is not clearly understood yet why gehlenite was precipitated in the column containing dead cells, while calcite was precipitated in the column with live cells.

3.2. Microscopic examination of microbiologically-induced CaCO_3 precipitation

Details of the consolidated column examined under SEM are depicted in Fig. 1(a)–(d), where crystals of distinct CaCO_3 precipitates grown between sand grains are clearly observed along with embedded bacteria (Fig. 1c). Rod-shaped bacteria were prominent in all sediment samples and appeared fossilized as intact bacilli in the middle of the calcite crystals (Fig. 1d). The presence of crystalline calcite associated with the

bacteria suggests that bacteria served as nucleation sites during the mineralization process.

3.3. Biochemistry of microbiologically-induced CaCO_3 precipitation

Fig. 2 presents patterns of calcite precipitation, cell growth, ammonia production and pH at an initial concentration of 25.2 mM CaCl_2 in the presence of live and killed cells. Similar patterns of microbial CaCO_3 precipitation were observed at different concentrations (12.6 and 50.4 mM) of the initial Ca^{2+} (not included in this paper). The microbiological CaCO_3 precipitation began at $\text{pH } 8.3 \pm 1.0$ and was completed at 9.0, consolidating 98% of the initial concentration of Ca^{2+} . Calcium carbonate precipitation appeared to be correlated with the growth of *B. pasteurii* and was completed within 16 h following inoculation. A considerable amount of ammonia was produced even during the stationary phase of cell growth. The pH of the medium also increased slowly as ammonia pro-

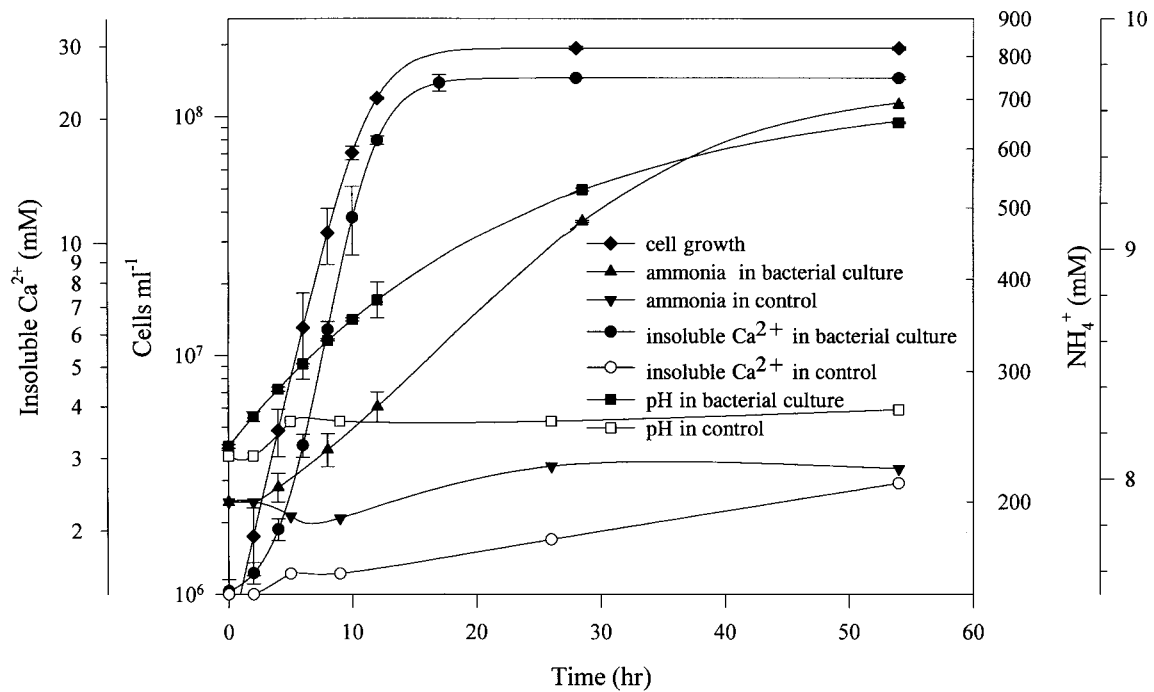


Fig. 2. Microbiologically induced CaCO_3 precipitation in the presence of *B. pasteurii* (inoculum size, 1.0×10^6 cells ml^{-1}) at 25.2 mM of CaCl_2 at 25°C. Each point represents the average of duplicate assays.

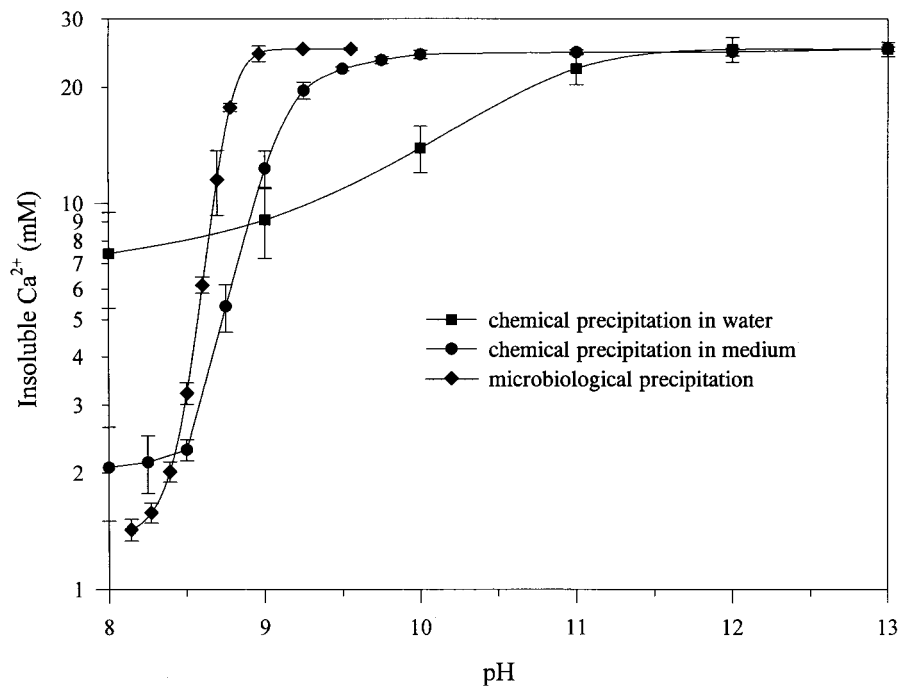


Fig. 3. Relationship between pH and CaCO_3 precipitation induced chemically and microbiologically. All experiments were carried out with 25.2 mM of CaCl_2 and NaHCO_3 , respectively, at 25°C. Each point represents the average of duplicate assays.

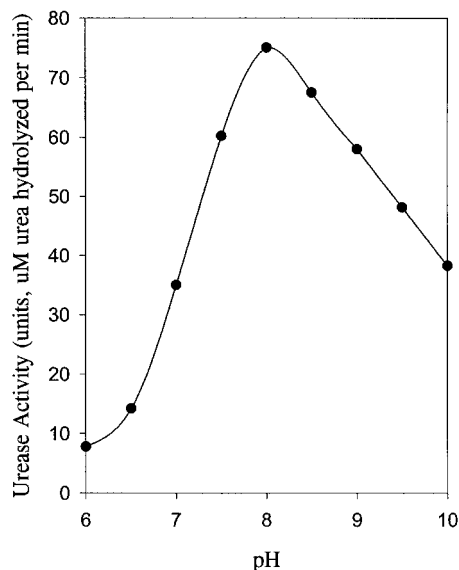


Fig. 4. Effect of pH on urease activity from the cell-free extract of *B. pasteurii*.

duction increased, but did not directly increase with the growth of cells. In the presence of killed cells, there were no detectable changes in calcite precipitation, ammonia production and pH.

3.4. Comparison of microbiologically-induced and chemically-induced CaCO_3 precipitation

To understand the effects of environmental factors, such as pH and ionic strength, patterns of CaCO_3 precipitation in the absence of microorganisms were examined with changes in pH in the water and in urea- CaCl_2 medium. In Fig. 3, the chemically-induced CaCO_3 precipitations in water and in medium at pHs ranging from 8 to 13 are compared with the microbiologically-induced CaCO_3 precipitation. The total amount of chemically-induced insoluble CaCO_3 increased as a function of pH in both water and medium. However, chemical CaCO_3 precipitation in medium was less at lower pHs and more at higher pHs than in water. While 98% of the initial Ca^{2+} concentration was precipitated microbiologically at pH 9.0, approximately 30 and 54% were precipitated chemically in water and in medium, respectively.

3.5. Urease activity and its Michaelis–Menten kinetic values

Urease activity was detected in the cell free extract of *B. pasteurii* at various pHs. However, no enzyme activity was detectable in the filtrate of *B. pasteurii* culture medium. Fig. 4 shows the effect of pH on urease activity from the cell free extract. As pH in the reaction mixture increased from 6.0 to 10.0, the enzyme ac-

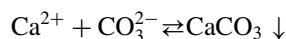
tivity increased at a relatively fast rate, peaking at pH around 8.0, and then decreasing slowly at higher pHs. Substantial amount of urease activity still remained at pH 9.0.

The Michaelis–Menten constants for urease from the cell extract were determined with varying concentrations of urea. The Michaelis–Menten kinetic parameters are particularly important in estimating enzymatic activities, in which K_m represents the substrate concentration at which the initial reaction rate is half maximal and $V_{\max}$ represents the maximum rate. From the Lineweaver–Burk plot, K_m and $V_{\max}$ values for urease from *B. pasteurii* at pH 7.0 were estimated to be 41.6 and 3.55 $\text{mM min}^{-1} \text{mg}^{-1}$, respectively. When the reaction mixture was adjusted to pH 7.7, the kinetic constants decreased to 26.2 mM for K_m and 1.72 $\text{mM min}^{-1} \text{mg}^{-1}$ for $V_{\max}$, demonstrating a higher affinity of the enzyme for urea at increased pH.

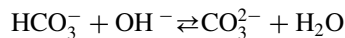
4. Discussion

4.1. Factors involved in microbiological CaCO_3 precipitation

The solubility of CaCO_3 is affected by ionic strength in the aqueous medium (Svehla, 1979; Stumm and Morgan, 1981). In calcite precipitation, the overall equilibrium reaction is



At 25°C, the solubility of CaCO_3 is estimated to be $3.8 \times 10^{-9} \text{ mol l}^{-1}$ of water at zero ionic strength, which increases to $6.3 \times 10^{-7} \text{ mol l}^{-1}$ at ionic strength equivalent to seawater. Carbonate ion is produced in water by the following equilibrium reactions:



In urea- CaCl_2 medium, NH_4^+ and Cl^- react with OH^- and H^+ , respectively, at different pH. Consequently, these reactions interfere with chemically-induced CaCO_3 precipitation as shown in Fig. 3, which indicates that CaCO_3 is more soluble in urea- CaCl_2 medium than in water at lower pHs, but reverse at higher pHs.

Microbiologically-induced CaCO_3 precipitation is recognized as a far more complicated process than chemically-induced precipitation. The bacterial cell surface with a variety of ions could nonspecifically induce mineral deposition by providing a nucleation site (Ferris et al., 1986, 1987). Especially, Ca^{2+} is not likely utilized by microbial metabolic processes, it

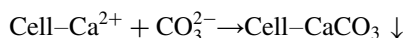
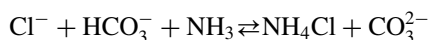
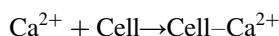
Table 2

Rate constants involved in microbiologically-induced precipitation of CaCO_3 at the different initial concentrations of CaCl_2 ^a: CaCO_3 precipitation rate, μ (h^{-1}); ammonia production rate, k (h^{-1})

Rate constant	Initial concentration of Ca^{2+} (mM)		
	12.6	25.2	50.4
μ (h^{-1})	0.76 (0.03)	0.77 (0.04)	0.72 (0.02)
k (h^{-1})	0.10 (0.01)	0.10 (0.03)	0.13 (0.01)

^a Inoculum: 1.0×10^6 cells ml^{-1} . (): standard deviation.

rather accumulates outside the cell (Silver et al., 1975). In medium, it is possible that individual microorganisms produce ammonia as a result of enzymatic urea hydrolysis to create an alkaline micro-environment around the cell. The high pH of these localized areas, without a significant increase in pH in the entire medium at the beginning, apparently commenced the growth of CaCO_3 crystals around the cell. Possible biochemical reactions in urea– CaCl_2 medium to precipitate CaCO_3 at the cell surface can be summarized as follows:



In culture medium, additional CO_2 is produced as

results of microbial respiration and enzymatic urea hydrolysis. The dissolved CO_2 , as a form of CO_3^{2-} or HCO_3^- , not only precipitates as part of calcite but also functions as buffer slowing down the pH increase as ammonia is produced from urea hydrolysis (Fig. 2).

4.2. Kinetics of microbiologically-induced CaCO_3 precipitation

As presented in Fig. 2, CaCO_3 precipitation is correlated to growth of bacteria and indirectly to production of ammonia. It is imperative to examine the kinetics of calcite precipitation as well as its relation to cell growth. The kinetics of calcite precipitation followed a modified logistic curve (Marquardt, 1963). The rate constants, μ for CaCO_3 formation and k for ammonia production, were calculated from regression analysis using an exponential logistic equation: $y = (a / (1 + e^{-b(x-c)})) + d$, where a is the range of y variation, $[\text{Ca}^{2+}]$ or $[\text{NH}_4^+]$; b is μ or k ; x is time (h); c is the time at the maximum (dy/dt) and d is the initial concentration of Ca^{2+} or NH_4^+ at time zero. Table 2 summarizes the values of μ and k at 12.6, 25.2 and 50.4 mM CaCl_2 . The rate constants at three different concentrations of CaCl_2 remained virtually the same, suggesting that rate constants, μ and k , are independent of the initial concentrations of Ca^{2+} added. The average rate of CaCO_3 precipitation is 0.75 h^{-1} and that of ammonia production is 0.11 h^{-1} .

Specific rates of CaCO_3 precipitation and ammonia production in relation to cell concentration were calcu-

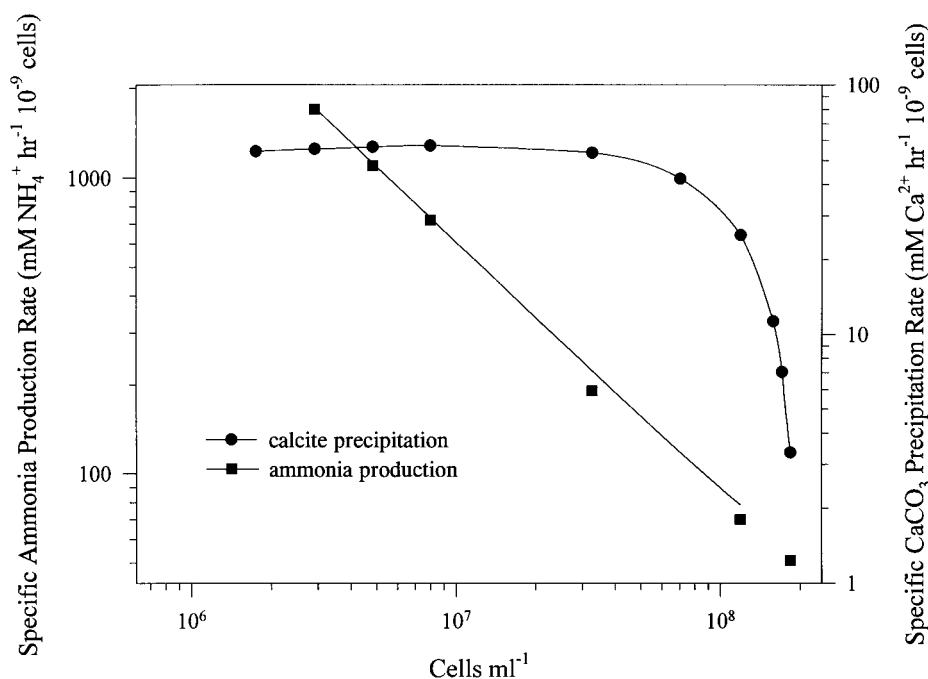


Fig. 5. Specific rates for CaCO_3 precipitation and ammonia production in relation to the cell concentration.

lated and graphed on a full logarithmic scale (Fig. 5). Two distinctly different curves are observed: a curved line for the CaCO_3 precipitation and a linear line for the ammonia production. The specific rate of CaCO_3 precipitation remained constant until the cell concentration became relatively high at which time the rate decreased considerably. However, the specific rate for ammonia production steadily decreased with increased cell concentration. It was hypothesized that, when the cell concentration was low, every cell would be responsible for serving as a nucleation site of CaCO_3 formation and producing ammonia to increase the pH in their immediate surroundings. Once the initial calcite precipitates on the nucleation site, calcite crystals continue to grow mainly due to the presence of microbial activities: increase in pH, production of carbon dioxide, and binding of Ca^{2+} to the cell surface (Morita, 1980). As calcite matrices grow thicker, urea becomes less available to the cells limiting further ammonia production.

4.3. Urease activity in *B. pasteurii*

Microbial urease is generally known to have a higher K_m , ranging between 9.2 and 230 mM, compared to the values of eucaryotic urease (Mörsdorf and Kaltwasser, 1989; Scott and Marguire, 1990; Christians et al., 1991; Evans et al., 1991). Variations in K_m values for urease ranging over 2 orders of magnitude have also been observed from plants and other microorganisms (Lynn, 1967; Smith et al., 1993; Natarajan, 1995). Kinetic values estimated for urease from *B. pasteurii* in our study are within the range reported from previous studies (Larson and Kallio, 1954; Mörsdorf and Kaltwasser, 1989; Ciurli et al., 1996).

Earlier, Kantzas et al. (1992) reported that urease from *B. pasteurii* was detected in the culture medium with a K_m value of 722.6 mM. They concluded that the *B. pasteurii* urease was extracellular and offered an option of using the urease rather than the whole cell to consolidate sand with CaCO_3 . In our study, the urease activity was identified in the cell crude extract, not in the filtrate of the culture medium containing growing cells, indicating that urease produced by *B. pasteurii* was likely cell-associated. It is, however, possible that some of the enzyme molecules might have leaked out of the cell and became extracellular, as determined by Kantzas et al. (1992). An increase in ammonia production during the stationary phase of cell growth (Fig. 2) also hinted a possibility that the urease was still active in degrading urea.

B. pasteurii is known to produce a large amount of urease in soil environments (Ciurli et al., 1996). The range of optimum pH values observed in our work is consistent with values reported by Larson and Kallio

(1954), Lynn (1967) and Evans et al. (1991). As indicated in Fig. 2, microbiologically-induced mineral consolidation occurs at pHs ranging between 8.3 and 9.0, where urease activity remains high. Thus, with a moderate enzymatic affinity (K_m of 26.2–41.6 mM) identified from our study, *B. pasteurii* urease would produce a sufficient amount of ammonia to maintain a high pH in its surroundings if an adequate amount of urea or related nitrogenous compounds were available.

In summary, observations from the XRD analysis and electron micrography have added credence to our previous studies (Gollapudi et al., 1995; Zhong and Islam, 1995), providing conclusive evidence of direct participation of *B. pasteurii* in calcite formation. Undoubtedly, *B. pasteurii* not only provides a nucleation site for calcite precipitation but also creates an alkaline environment inducing further growth of calcite. Results of kinetic studies render an explanation that the rate of CaCO_3 precipitation correlates with cell growth and urease exhibits higher enzymatic activities and stronger affinity to urea at higher pH values (pH 8–9) where the calcite precipitation is favorable. Our findings further support the notion that in-situ implementation of microbial mineral plugging might provide a practical means for bioremediation of porous media. Currently, an immobilization technique that encapsulates *B. pasteurii* in the matrix of polyurethane is being investigated to induce a large quantity of calcite precipitation in remediation of concrete cracks.

Acknowledgements

This research was funded by a grant from the National Science Foundation. We express our sincere appreciation to Dr. M.R. Islam and Dr. V. Ramakrishnan who have provided helpful comments and suggestions for the research. We also acknowledge Dr. E.F. Duke and Dr. B.L. Davis of the Engineering and Mining Experiment Station at the South Dakota School of Mines and Technology for their technical assistance in SEM and XRD. Special thanks go to D. Johnston of the South Dakota Department of Transportation for his critical review of kinetic data.

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